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Low-noise amplifier (LNA) with high power supply rejection ratio (PSRR)

Abstract

A low-noise amplifier includes a low-noise amplifier stage and a filtering and biasing stage. The low-noise amplifier stage receives an input signal and provides a first output signal in response thereto. The low-noise amplifier stage includes a gain element for proving the first output signal, and at least one lowpass filter circuit in series between a first power supply voltage terminal and the gain element having a conductivity determined by lowpass filtering a signal at a bias terminal, and a filtering and biasing stage having an input for receiving the first output signal, and an output for providing a second output signal, and at least one cascode element having a first current conduction path coupled in series between the bias terminal and the output, and having a predetermined filter characteristic.

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/260,464, filed on Aug. 20, 2021, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to amplifier circuits, and more specifically to low noise amplifiers for use in applications such as wireless communication systems.

BACKGROUND

(2) A low-noise amplifier (LNA) is an electronic amplifier that amplifies a very low-power signal

without significantly degrading its signal-to-noise ratio. Low-noise amplifiers (LNA) are common blocks in wireless communication systems, such as near-field magnetic induction (NFMI) receivers, radio frequency (RF) receivers, etc. For lowest noise and highest energy efficiency, a single-ended implementation is generally preferred.
(3) The performance of LNA is dependent not only on the design of the amplifier itself, but often also by external factors such as power supply noise, especially for single-ended designs. While very high frequency power supply noise in the giga-Hertz (GHz) range can be relatively easily filtered by passive devices, it becomes more difficult to filter out power supply noise in lower frequency ranges, such as in the mega-Hertz (MHz) range. As a result, it has been difficult to exploit the low-noise and high-efficiency capabilities of single-ended LNAs, and fully-differential LNAs are often used instead.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings, in which:
(2) FIG. **1** illustrates in schematic form a conventional low noise amplifier (LNA) known in the prior art;
(3) FIG. **2** illustrates in schematic form a low noise amplifier (LNA) with a high power-supply rejection ratio (PSRR) according to various embodiments of the present disclosure; and
(4) FIG. **3** illustrates in schematic form a portion of a filtering stage of a low noise amplifier including a high pass filter having a high pass characteristic according to various embodiments of the present disclosure.
(5) The use of the same reference symbols in different drawings indicates similar or identical items. Unless otherwise noted, the word “coupled” and its associated verb forms include both direct connection and indirect electrical connection by means known in the art, and unless otherwise noted any description of direct connection implies alternate embodiments using suitable forms of indirect electrical connection as well.
DETAILED DESCRIPTION
(6) FIG. **1** illustrates in schematic form a conventional low noise amplifier (LNA) **100** known in the prior art. LNA **100** includes generally an input stage **110**, a biasing stage **120**, and an output stage **130**.
(7) Input stage **110** includes generally transistors **111-113** and a current mirror **114**. Transistor **111** is a P-channel metal-oxide-semiconductor (MOS) transistor having a source connected to a power supply voltage terminal labelled “V.sub.DD”, a gate, and a drain. V.sub.DD is a more-positive power supply voltage terminal having a nominal voltage of, for example, 3.0 volts. Transistor **112** is a P-channel MOS transistor having a source connected to the drain of transistor **111**, a gate for receiving an input signal labelled “V.sub.IN+”, and a drain providing an output of input stage **110**. Transistor **113** is a P-channel MOS transistor having a source connected to the drain of transistor **111**, a gate for receiving an input signal labelled “V.sub.IN−”, and a drain. Current mirror **114** includes transistors **115** and **116**. Transistor **115** is an N-channel MOS transistor having a drain connected to the drain of transistor **113**, a gate connected to the drain thereof, and a source connected to a power supply voltage terminal labelled “V.sub.SS”. V.sub.SS is a more-negative or ground power supply voltage terminal having a nominal voltage of 0.0 volts. Transistor **116** is an N-channel MOS transistor having a drain connected to the drain of transistor **112**, a gate connected to the drain of transistor **115**, and a source connected to V.sub.SS.
(8) Biasing stage **120** includes transistors **121** and **122**. Transistor **121** is a P-channel MOS transistor having a source connected to V.sub.DD, a gate connected to the gate of transistor **111**,

and a drain connected to the gate thereof. Transistor **122** is an N-channel MOS transistor having a drain connected to the drain of transistor **121**, a gate connected to the drain of transistor **112**, and a source connected to V.sub.SS.

(9) Output stage **130** includes a resistor **131**, and a transistor **132**. Resistor **131** has a first terminal connected to V.sub.DD, and a second terminal providing an output of LNA **100** labelled "V.sub.OUT". Transistor **132** is an N-channel MOS transistor having a drain connected to the second terminal of resistor **131**, a gate connected to the drain of transistor **112**, and a source connected to V.sub.SS.

(10) In operation, transistor **111** generates a bias current for input stage **110**, and transistors **112** and **113** divert the bias current based on the difference between voltages VIN+ and VIN−, which changes the voltage on the drain of transistor **112** in response thereto. Current mirror **114** mirrors the current flowing through transistor **113** to the drain-to-source current of transistor **116** based on the gate width-to-gate length ratio of transistors **115** to **116**. This current is in turn mirrored by a current mirror formed by transistors **121** and **111** to form the source-to-drain current of transistor **111** based on the gate width-to-gate length ratio of transistors **121** to **111**. When the currents balance, the voltage on the drain of transistor **112** is proportional to the difference in voltage between V.sub.IN+ and V.sub.IN−. The voltage controls the conductivity of transistor **132** such that the drain-to-source current of transistor **132** varies linearly with small-signal changes between V.sub.IN+ and V.sub.IN−. These linear changes affect the current-resistance (IR) drop across resistor **131**, and thus V.sub.OUT varies proportionately to the difference between V.sub.IN+ and V.sub.IN−.

(11) LNA **100** is a typical differential input operational amplifier having a self-current biasing structure, and many other known LNAs are built on the basic structure of LNA **100**. Even with the differential input structure, its PSRR is limited to about 40 dB. An LNA architecture that provides better PSRR would be desirable.

(12) FIG. **2** illustrates in schematic form a low noise amplifier (LNA) **200** with a high PSRR according to various embodiments of the present disclosure. LNA **200** includes generally a low-noise amplifier stage **210**, a filtering and biasing stage **250**, and one or more optional filtering stages **260**.

(13) Low-noise amplifier stage **210** includes transistors **211-214**, a resistor **220**, a lowpass filter circuit **230**, and a lowpass filter circuit **240**. Transistor **211** is an N-channel MOS transistor having a drain, a gate for receiving an input signal labelled "IN", and a source connected to ground. Transistor **212** is a P-channel MOS transistor having a source, a gate for receiving a control signal labelled "NEN", and a drain connected to the drain of transistor **211**. Transistor **213** is an N-channel MOS transistor having a drain, a gate, a source connected to the source of transistor **212**, and a bulk terminal connected to ground. Transistor **214** is an N-channel MOS transistor having a drain for receiving a power supply voltage labelled "VLOW", a gate, a source connected to the drain of transistor **213**, and a bulk terminal connected to the gate thereof. Resistor **220** has a first terminal connected to the drain of transistor **211**, and a second terminal connected to the gate of transistor **211**. Lowpass filter **230** includes a resistor **231**, and a capacitor **232**. Resistor **231** has a first terminal connected to the bias node, and a second terminal connected to the gate of transistor **213**. Capacitor **232** has a first terminal connected to the second terminal of resistor **231** and to the gate of transistor **213**, and a second terminal connected to ground. Lowpass filter **240** includes a resistor **241**, and a capacitor **242**. Resistor **241** has a first terminal connected to the bias node, and a second terminal connected to the gate of transistor **214**. Capacitor **242** has a first terminal connected to the second terminal of resistor **241** and to the gate of transistor **214**, and a second terminal connected to ground.

(14) Filtering and biasing stage **250** includes transistors **251-254** and a bandpass network **255**. Transistor **251** is an N-channel MOS transistor having a drain, a gate, and a source connected to ground. Transistor **252** is a P-channel MOS transistor having a source, a gate for receiving a control

signal labelled “NEN_BIAS”, and a drain connected to the drain of transistor **251**. Transistor **253** is an N-channel MOS transistor having a drain, a gate, and a source connected to the source of transistor **252**, and a bulk terminal connected to ground. Transistor **254** is an N-channel MOS transistor having a drain for receiving a power supply voltage labelled “VLOW”, a gate, a source connected to the drain of transistor **253**, and a bulk terminal connected to the bias node. Bandpass network **255** includes a resistor **256**, a variable capacitor **257**, a variable capacitor **258**, and a resistor **259**. Resistor **256** has a first terminal connected to the drains of transistors **211** and **212**, and a second terminal. Variable capacitor **257** has a first terminal connected to the second terminal of resistor **256**, a second terminal connected to the drains of transistors **251** and **252**, and a tuning input. Variable capacitor **258** has a first terminal connected to the second terminal of resistor **256**, a second terminal connected to the gate of transistor **251**, and a tuning input. Resistor **259** has a first terminal connected to the second terminal of variable capacitor **257** and to the drains of transistors **251** and **252**, and a second terminal connected to the gate of transistor **251**.

(15) In various embodiments, each of one or more optional filtering stages **260** has the same construction besides the exemplary filtering stage **260** shown in FIG. **2**. Filter stage **260** includes transistors **261-264** and a bandpass network **265**. Transistor **261** is an N-channel MOS transistor having a drain, a gate, and a source connected to ground. Transistor **262** is a P-channel MOS transistor having a source, a gate for receiving control signal NEN, and a drain connected to the drain of transistor **261**. Transistor **263** is an N-channel MOS transistor having a drain, a gate, and a source connected to the source of transistor **262**, and a bulk terminal connected to ground. Transistor **264** is an N-channel MOS transistor having a drain for receiving power supply voltage VLOW, a gate connected to the bias node, a source connected to the bias node, and a bulk terminal connected to the bias node. Bandpass network **265** includes a resistor **266**, a variable capacitor **267**, a variable capacitor **268**, and a resistor **269**. Resistor **266** has a first terminal connected to the drains of transistors **251** and **252**, and a second terminal. Variable capacitor **267** has a first terminal connected to the second terminal of resistor **266**, a second terminal connected to the drains of transistors **261** and **262**, and a tuning input. Variable capacitor **268** has a first terminal connected to the second terminal of resistor **266**, a second terminal connected to the gate of transistors **261**, and a tuning input. Resistor **269** has a first terminal connected to the second terminal of variable capacitor **267** and to the drains of transistors **261** and **262**, and a second terminal connected to the second terminal of capacitor **268** and to the gate of transistor **261**.

(16) In operation, LNA **200** achieves triple-stage isolation with a low-voltage bias scheme in order to deliver amplification with a very high PSRR and filtering according to a desired filter characteristic. Filtering and biasing stage **250** includes an amplification element, i.e., transistor **251**, in series with three transistors, including a P-channel power off transistor **252** in series with an input leg of a current mirror formed by two N-channel transistors **253** and **254**. Transistor **252** disables filtering and biasing stage **250** when control signal NEN_BIAS is inactive at a logic high, and enables filtering and biasing stage **250** when control signal NEN_BIAS is active at a logic low. When enabled, filtering and biasing stage **250** forms the input side of a current mirror by setting the voltage at the bias node.

(17) LNA stage **210** and each filtering stage **260** include an amplification element in series with three transistors, in which two of the three transistors form an output leg of the current mirror, and the third operates as a switch in conduction. In LNA stage **210**, transistor **211** forms the amplification element, and its DC bias point is set by resistor **220**. Transistors **213** and **214** form an output leg of the current mirror, but LNA stage **210** uses lowpass filters **230** and **240** to filter the bias node to provide a filtered bias voltage to the gates of current source transistors **214** and **215**. Since LNA stage **210** is the most noise sensitive stage, lowpass filters **230** and **240** are added to filter the gate voltages of transistors **213** and **214** to increase the PSRR. Forward body biasing is used on transistor **214** by connecting the body electrode to the gate electrode and thereby to set the drain-to-source voltage of transistor **213**. By using N-channel MOS transistors for the current

source, LNA stage **210** refers the filtering capacitors **232** and **242** to ground, which is the reference of the input signal. In each filtering stage **260**, transistor **261** forms the amplification element, and its DC bias point is set by resistor **269**. Transistors **263** and **264** form another output leg of the current mirror, but filtering stage **260**. Forward body biasing is used on transistor **264** by connecting the body electrode to the gate electrode and thereby to set the drain-to-source voltage of transistor **263**.

(18) The bias generation mechanism relies on an N-channel MOS current mirror stacked between the positive power supply voltage and the active LNA circuit, thus providing a very low supply voltage requirement for the active circuit. The bias generation mechanism for this current mirror relies on a current reuse method in which the branch current, which is used to generate the bias voltage, is also utilized for the amplification by embedding the bias generation and LNA stage in the same stack of devices. Therefore, LNA **200** causes near-zero wasted current and increases the power efficiency of LNA **200** while providing superior power supply rejection. The same basic structure is used for a very efficient implementation of a bandpass filter by using second order band-pass filters whose bandpass characteristic can be adjusted by tuning the capacitors.

(19) LNA **200** achieves a very high PSRR, e.g., greater than 80 dB, by using a stack of two NMOS current sources on the output sides of a current mirror of which the input side is formed by corresponding transistors in filtering and biasing stage **250**.

(20) LNA **200** also achieves low power because the bias current is drawn from a low-voltage power supply voltage terminal with almost no current wasted and the only current is devoted in an amplification device, i.e., transistor **211**. In an exemplary embodiment using modern CMOS transistor technology, VLOW can be set at 0.9V.

(21) Moreover, LNA **200** provides low noise in a single-ended signaling arrangement. It uses a simple amplification element, i.e., N-channel MOS transistor **211** in FIG. **2**.

(22) An integrated power-off element, e.g., transistor **212** in LNA stage **210**, provides fast startup, such as less than one microsecond (p).

(23) In addition, LNA **200** makes it easy to implement a higher-order filter function with the addition of filtering stages. For example, LNA **200** could implement a second-order bandpass filter by adding a single filtering stage **260**. Higher-order bandpass filters can be achieved by adding one or more filtering stages **260**.

(24) FIG. **3** illustrates in schematic form a portion of a filtering stage **300** of a low noise amplifier including a high pass filter **355** according to various embodiments of the present disclosure. LNA **300** includes a transistor **351** and high pass filter **355**. Transistor **351** is an N-channel MOS transistor having a drain providing an output of filtering stage **300**, a gate, and a source connected to ground. High pass filter **355** includes a variable capacitor **356**, and a resistor **357**. Variable capacitor **356** has a first terminal connected to the output of the previous stage, a second terminal connected to the gate of transistor **351**, and a tuning input. Resistor **357** has a first terminal connected to the second terminal of variable capacitor **357** and to the gate of transistors **351**, and a second terminal connected to the drain of transistor **351**.

(25) When filtering and biasing stage **250** of LNA **200** of FIG. **2** exhibits a high pass characteristic, then transistor **351** can be used in place of transistor **251** and high pass filter **355** can be used in place of bandpass filter **255**. Similarly, transistor **351** can be used in place of transistor **261** and high pass filter **355** can be used in place of bandpass filter **265** in each filtering stage **260**. Therefore, LNA **300** of FIG. **3** has the same advantages as LNA **200** of FIG. **2** except that it has integral high pass filter characteristics.

(26) The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments that fall within the scope of the claims. For example, an LNA as disclosed herein can include stages having either integral bandpass filter characteristics or integral high pass filter characteristics. The order of the filter can be increased by using additional filtering stages. The low

noise amplifier stage can have only one transistor with a corresponding lowpass filter, or more than two.

(27) Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the forgoing detailed description.

Claims

1. A low-noise amplifier, comprising: a low-noise amplifier stage for receiving an input signal and providing a first output signal in response thereto, said low-noise amplifier stage comprising: a gain element for providing said first output signal; and at least one lowpass filter circuit in series between a first power supply voltage terminal and said gain element and having a conductivity determined by lowpass filtering a signal at a bias terminal, and a filtering and biasing stage having an input for receiving said first output signal, and an output for providing a second output signal, and at least one cascode element having a first current conduction path coupled in series between said bias terminal and said output, and having a predetermined filter characteristic.
2. The low-noise amplifier of claim 1, wherein said predetermined filter characteristic comprises a bandpass characteristic.
3. The low-noise amplifier of claim 2, further comprising: at least one bandpass stage having an input for receiving said second output signal, and an output for providing an output of the low-noise amplifier, each of said at least one bandpass stage having a respective cascode element coupled in series between said first power supply voltage terminal and a respective output, and having a bandpass characteristic.
4. The low-noise amplifier of claim 1, wherein said predetermined filter characteristic comprises a high pass characteristic.
5. The low-noise amplifier of claim 1, wherein each of said at least one lowpass filter circuit comprises: a transistor having a first current electrode, a control electrode, and a second current electrode, wherein said second current electrode of said transistor of a first one of said at least one lowpass filter circuit is coupled to said output of said gain element, and said first current electrode of said transistor of a last one of said at least one lowpass filter circuit is coupled to said first power supply voltage terminal; a resistor having a first terminal coupled to said bias terminal, and a second terminal coupled to said control electrode of said transistor; and a capacitor having a first terminal coupled to said control electrode of said transistor, and a second terminal coupled to a second power supply voltage terminal.
6. The low-noise amplifier of claim 1, wherein said low-noise amplifier stage further comprises an enable transistor coupled between said at least one lowpass filter circuit and said output of said gain element.
7. The low-noise amplifier of claim 6, wherein said bias terminal receives a bias current.
8. A low-noise amplifier, comprising: a low-noise amplifier stage having a first gain element with an input for receiving an input signal, and an output for providing a first amplified signal, and at least one lowpass filter circuit each having an input coupled to a bias terminal and a first current conduction path coupled between a first power supply voltage terminal and said output of said first gain element; a bandpass and biasing stage having a second gain element with an input coupled to said output of said first gain element, and an output for providing a second amplified signal, and at least one cascode element having an input coupled to said bias terminal, and a second current conduction path coupled in series between said bias terminal and said output of said second gain element; and at least one bandpass stage coupled in series, each having a respective gain element with an input coupled to an output of a previous bandpass stage, and an output for providing a corresponding amplified signal, and at least one corresponding cascode element having an input coupled to said bias terminal, and a respective current conduction path coupled in series between

said first power supply voltage terminal and said output of a corresponding previous gain element.
9. The low-noise amplifier of claim 8, wherein each of said first gain element, said second gain element, and said respective gain element of said at least one bandpass stage comprises an N-channel metal-oxide-semiconductor (MOS) transistor.
10. The low-noise amplifier of claim 8, wherein each of said at least one lowpass filter circuit comprises: a transistor having a first current electrode, a control electrode, and a second current electrode, wherein said second current electrode of said transistor of a first one of said at least one lowpass filter circuit is coupled to said output of said first gain element, and said first current electrode of said transistor of a last one of said at least one lowpass filter circuit is coupled to said first power supply voltage terminal; a resistor having a first terminal coupled to said bias terminal, and a second terminal coupled to said control electrode of said transistor; and a capacitor having a first terminal coupled to said control electrode of said transistor, and a second terminal coupled to a second power supply voltage terminal.
11. The low-noise amplifier of claim 8, wherein said low-noise amplifier stage comprises a first enable transistor coupled between said at least one lowpass filter circuit and said output of said first gain element.
12. The low-noise amplifier of claim 11, wherein said bandpass and biasing stage comprises a second enable transistor coupled between said at least one cascode element and said output of said second gain element, and each of said at least one bandpass stage comprises a respective enable transistor coupled between said at least one respective cascode element and said output of said respective gain element.
13. The low-noise amplifier of claim 11, wherein said bias terminal receives a bias current.
14. The low-noise amplifier of claim 8, wherein said bandpass and biasing stage further comprises a bandpass network coupled between said input and said output thereof.
15. The low-noise amplifier of claim 14, wherein said bandpass network further comprises: a first resistor having a first terminal coupled to said output of said first gain element, and a second terminal; a first capacitor having a first terminal coupled to said second terminal of said first resistor, and a second terminal coupled to said output of said second gain element; a second capacitor having a first terminal coupled to said second terminal of said first resistor, and a second terminal coupled to said input of said second gain element; and a second resistor having a first terminal coupled to said output of said second gain element, and a second terminal coupled to said input of said second gain element.
16. A method of amplifying an input signal to provide an output signal with a very high power-supply rejection ratio (PSRR), comprising: amplifying a radio-frequency (RF) input signal using a first gain element coupled between a first power supply voltage terminal and a second power supply voltage terminal and providing an amplified signal in response to said amplifying; biasing said first gain element, said biasing comprising modulating a conductivity of at least one cascode transistor, and said biasing further comprising: generating a bias voltage at a bias terminal using a bias current; lowpass filtering said bias voltage to form a lowpass filtered bias voltage; and biasing a control electrode of each of said at least one cascode transistor using said lowpass filtered bias voltage; and creating said bias voltage in response to said amplified signal.
17. The method of claim 16, wherein creating said bias voltage in response to said amplified signal comprises: bandpass filtering said amplified signal and forming a bandpass filtered amplified signal in response; modulating a conductivity of a second gain element in response to said bandpass filtered amplified signal, and generating a second amplified signal in response to said modulating; and biasing said second gain element using at least one cascode element having a control electrode coupled to said bias terminal, and a second current conduction path coupled in series between said bias terminal and a current conduction path of said second gain element.
18. The method of claim 17, further comprising: bandpass filtering said second amplified signal in at least one bandpass stage; and forming the output signal in response to said bandpass filtering.

19. The method of claim 16, wherein: amplifying said RF input signal using said first gain element comprises amplifying said RF input signal using an N-channel MOS transistor.
20. The method of claim 16, wherein modulating said conductivity of said at least one cascode transistor further comprises modulating a bulk terminal of an N-channel MOS transistor using said lowpass filtered bias voltage.